

Analog Lab Report Week 1

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Objective :

The objective of Week 1 is to derive the expressions for the V_{Ramp} and V_{Square} waves using operational amplifiers (op-amps). We use a capacitor in an integrator configuration to integrate the square wave and generate a ramp wave, and a comparator to compare V_{RAMP} with V_{CM} to produce the square wave.

List of Components :

1. $R_1 = 25k\Omega$, $R_2 = 40k\Omega$, $R_3 = 200k\Omega$
2. $C_1 = 10nF$
3. OPM1 (MCP6004) = 1
4. CMP1 (LM339) = 1
5. Pull up Resistor = $3.7k\Omega$
6. Power Supply (ADALM 1000)
7. Breadboard

EXPERIMENT-1: RAMP GENERATOR

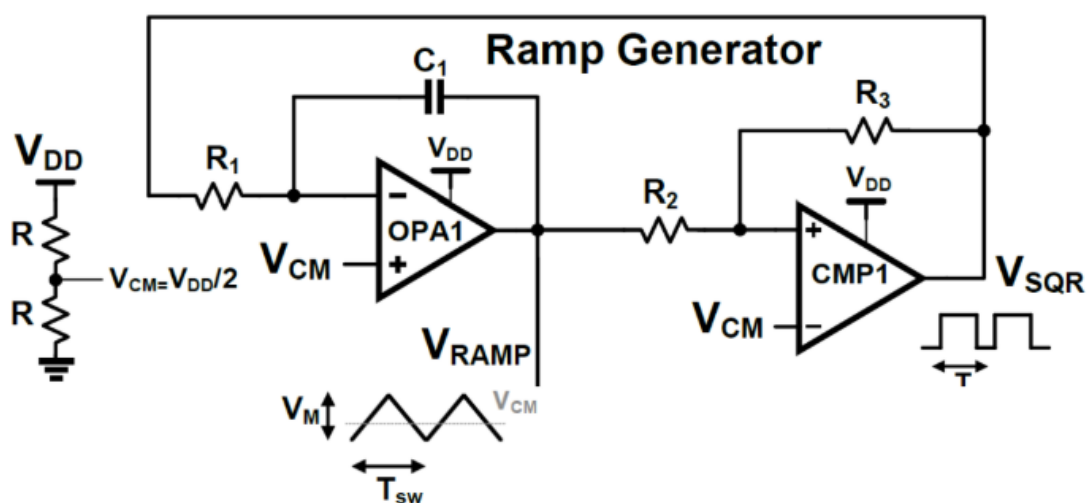


Figure 1 Schematic diagram of a ramp generator.

Derivation of Formula :

$$V_+ = \frac{R_3}{R_2 + R_3} V_{RAMP} + \frac{R_2}{R_2 + R_3} V_{SAR}$$

switching when

$$V_+ = V_{CM}$$

Case 1: when output is low ($V_{SAR} = 0$)

$$V_{CM} = \frac{R_3}{R_2 + R_3} V_{RAMP}$$

$$V_{RAMP,1} = \frac{R_2 + R_3}{R_3} V_{CM}$$

Case 2: when output is high ($V_{SAR} = V_{DD}$)

$$V_{CM} = \frac{R_3}{R_2 + R_3} V_{RAMP,H} + \frac{R_2}{R_2 + R_3} V_{DD}$$

$$\left\{ V_{CM} = V_{DD}/2 \right\}$$

$$V_{DD} \left(\frac{1}{2} - \frac{R_2}{R_2 + R_3} \right) = \frac{R_3}{R_2 + R_3} V_{RAMP,H}$$

$$V_{RAMP,H} = \frac{R_2 + R_3}{R_3} \left(\frac{1}{2} - \frac{R_2}{R_2 + R_3} \right) V_{DD}$$

$$V_{RAMP,H} = \left(\frac{1(R_2 + R_3)}{2 R_3} - \frac{R_2}{R_3} \right) V_{DD}$$

$$V_{RAMP, H} = \frac{R_2 + R_3}{R_3} V_{CM} - \frac{R_3}{R_3} V_{DD}$$

$$\begin{aligned} V_{PP} &= V_{RAMP, L} - V_{RAMP, H} \\ &= \frac{R_2 + R_3}{R_3} V_{CM} - \left(\frac{R_2 + R_3}{R_3} V_{CM} - \frac{R_2}{R_3} V_{DD} \right) \\ &= \frac{R_2}{R_3} \frac{V_{DD} \times 2}{2} - \left(\text{Equation 1} \right) \\ &\quad 2 V_{CM} \times R_2 / R_3 \end{aligned}$$

ramp amplitude depends on R_2/R_3 & supply voltage

for RAMP Slope (Integrator)

$$i_{R_1} = \frac{V_{CM} - V_{IOB}}{R_1}$$

0 as dc

$$i_{R_1} = \frac{d(V_{CM} - V_{RAMP})}{dt}$$

$$\frac{dV_{RAMP}}{dt} = \frac{V_{CM} - V_{IOB}}{R_1 C_1}$$

$$\frac{dV_{RAMP}}{dt} = \frac{V_{IOB} - V_{CM}}{R_1 C_1}$$

$$V_{SQR} - V_{CM} = \frac{-V_{DD}}{2}$$

when output is high:

$$V_{RAMP,2} = V_{SQR} - V_{CM} = \frac{+V_{DD}}{2}$$

$$V_{PP} = V_{RAMP,2} - V_{RAMP,1} = \frac{V_{DD}}{2} - \left(\frac{-V_{DD}}{2} \right) = V_{DD}$$

$$\frac{dV_{RAMP}}{dt} = \frac{V_{DD} - V_{DD}/2}{R_1 C_1}$$

$$\frac{dV_{RAMP}}{dt} = \frac{V_{DD}}{2R_1 C_1}$$

~~~~~ slope of RAMP

$$\text{Voltage diff} = \frac{V_{PP}}{5}$$

$$t_{half} = \frac{V_{DD} \cdot \frac{R_2}{R_3} \cdot \frac{V_{DD}}{2R_1 C_1}}{\frac{V_{DD}}{2R_1 C_1}} \Rightarrow \frac{R_2 \times 2R_1 C_1}{R_3}$$

Total period  $T = 2 \text{ half}$   
 $\cdot 4 R_1 C_1 R_2$   
 $R_3$

$$f = \frac{1}{T} = \frac{R_3}{4 R_1 C_1 R_2} \quad (2)$$

$$V_{DD} = 5V$$

$$f_{in} = 5 \text{ kHz}$$

$$V_M = 1V$$

$R_2/R_3 \rightarrow$  sets amplitude

$R_1 C_1 \& R_3/R_2 \rightarrow$  sets frequency

$$V_M = \frac{2 R_2}{R_3} V_{DD}$$

$$1 = \frac{R_2}{R_3} 5$$

$$\boxed{R_3 = 5 R_2}$$

$$f_{in} = \frac{R_3}{4 R_1 C_1 R_2}$$

$$\frac{1}{5 \text{ kHz}} = \frac{1}{5 \cancel{R_2}} \cdot \frac{1}{4 R_1 C_1 \cancel{R_2}}$$

$$R_1 C_1 = \frac{1}{4 \text{ kHz}}$$

$$R_1 C_1 = 250 \mu\text{s} \quad \text{or } 25 \text{ ms}$$

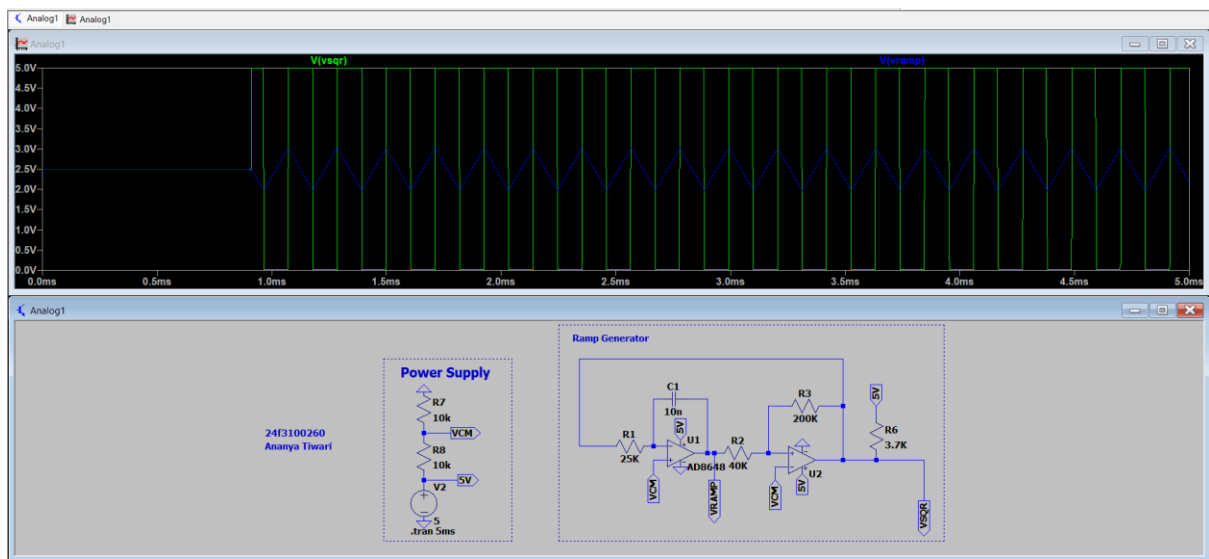
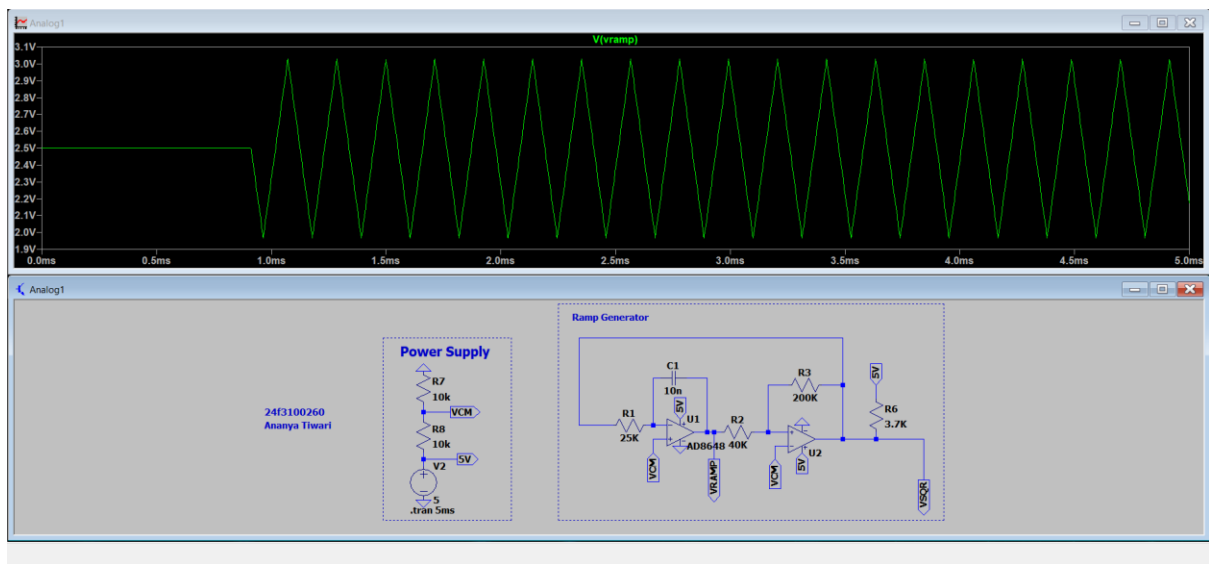


if select  $C_1 = 10\text{mF}$  and solve for  $R$

$$R_1 = \frac{0.25\text{ms}}{10\text{mF}} \Rightarrow \frac{25 \times 10^{-5}\text{s}}{10^{-3}\text{F}}$$

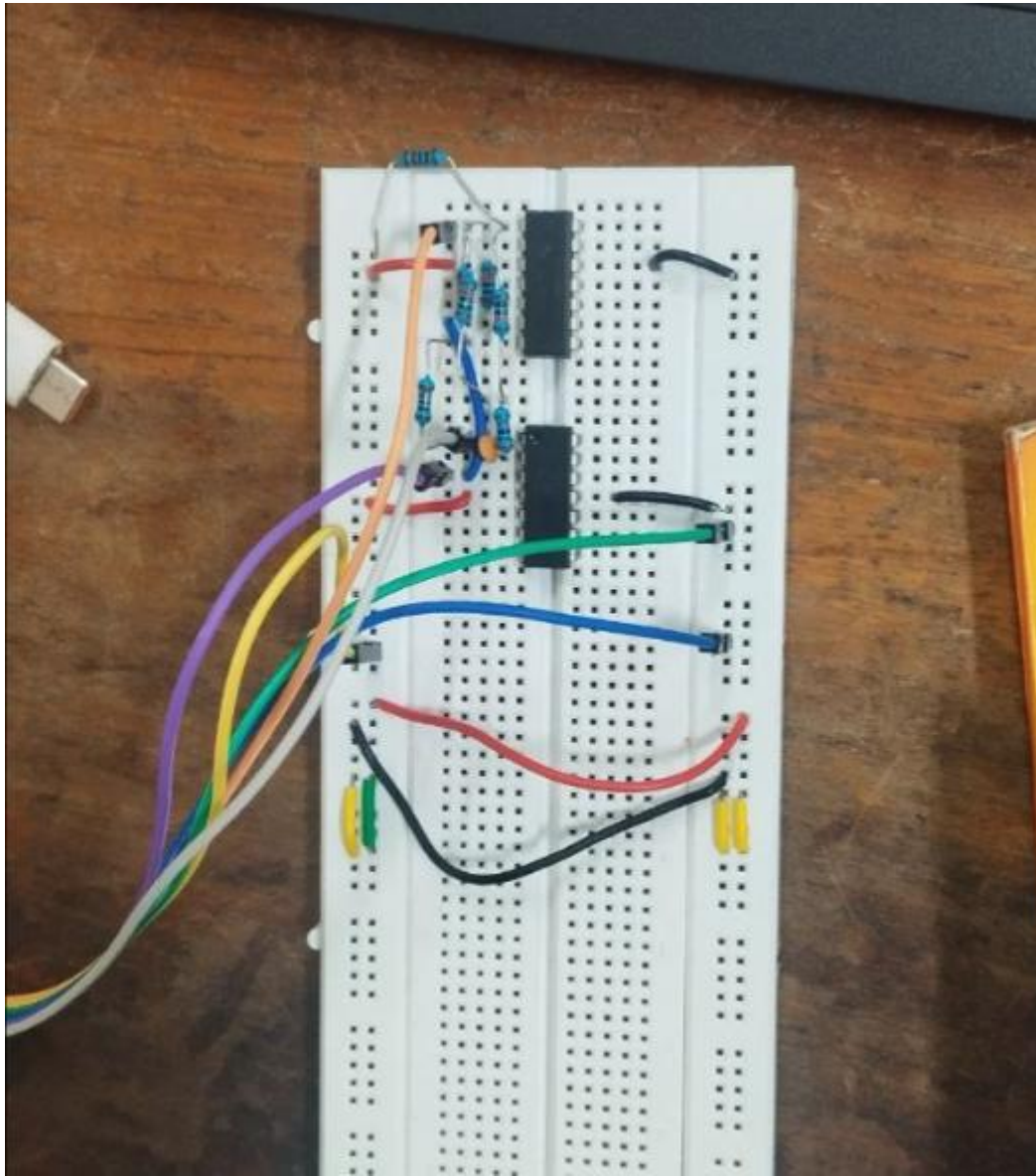
$$\Rightarrow 25\text{k}\Omega$$

### LTspice Schematics and Output :



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### Physical Circuit :



### Physical Circuit Wire Connections :-

Green wire and Blue wire – GND

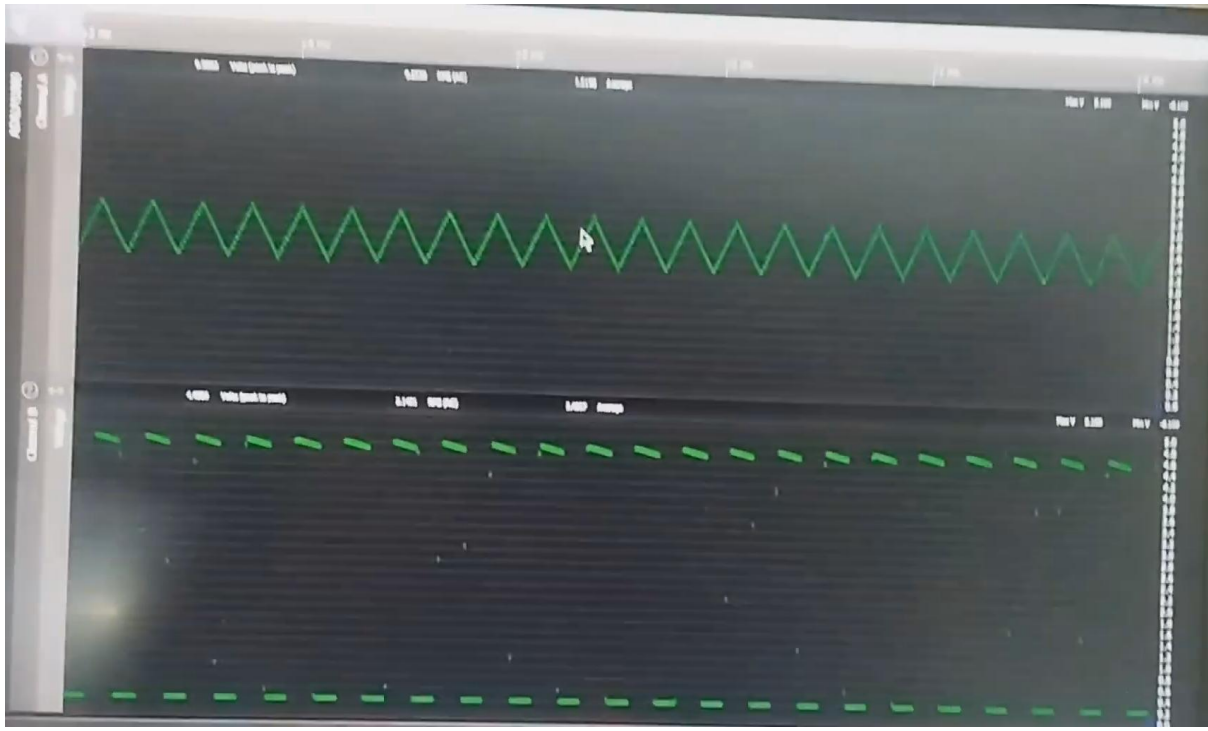
Yellow – 5V

Orange wire – Channel B ( $V_{Square}$ )

White wire – Channel A ( $V_{Ramp}$ )

Purple wire – 2.5V

### Physical Circuit Output :-



The peak-to-peak voltage of the ramp deviates from 1V because  $V_{DD}$  and  $V_{CM}$  vary from their respective 5V and 2.5V targets; the ADALM1000 simply lacks the capacity to deliver that specific voltage level.

#### Conclusion :-

The experiment was successful, with the physical hardware output validating the simulation models.